

Learned Z-Score Forecasting for Cross-Exchange Cryptocurrency Spreads: A Gradient-Boosting Approach on Live Microstructure Data

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Abstract

Identical cryptocurrencies often trade at different prices across exchanges. Most high-frequency pairs strategies still turn that gap into a trade with a simple rule: enter when the rolling z-score of the spread is large, and wait for mean reversion. We show that a better signal is to *predict* the next z-score. Using LightGBM on live orderbook imbalance, trade flow, and lagged spread dynamics—including L2 signals that cannot be rebuilt from historical candle APIs—we forecast the next-snapshot cross-exchange z-score for the same asset across venues, and trade only when the forecast is confidently large ($|\hat{z}| \geq 0.9$). We choose this gate as the first threshold where per-trade Sharpe reaches ≥ 1 while retaining high mean pn1_proxy without cutting the book as aggressively as stricter neighbors. We test the policy out of sample in live paper trading on real market data, in a short Jul 31 run (~ 8 hours) and a longer ~ 73 -hour campaign at roughly one-minute resolution. Trades are scored in z-score units, not dollars: a trade is a win if the realized next z-score moves in the predicted direction, and the per-trade score is $\text{pn1_proxy} = \text{sign}(\hat{z}) \times z_{\text{exit}}$. Under the $|\hat{z}| \geq 0.9$ protocol on the multi-day book, about 87% of settled trades are directional wins, with mean pn1_proxy of +1.37 across 12,795 closes. The Jul 31 session shows the same pattern (DirAcc 84.9%; prediction R^2 of 0.51 on $\tau=0.9$ entries) and, under matched capacity at the same gate, beats mechanical z-score persistence by 15.4 percentage points in win rate and 63% in mean pn1_proxy . We publicly release the code repository, trained model, and synchronized multi-exchange microstructure dataset (including orderbook, trades, funding, and open interest) on Hugging Face (the code repository (URL withheld for double-blind review); the public dataset release (URL withheld for double-blind review)) so others can reproduce and extend the work.

CCS Concepts

• **Computing methodologies** → **Supervised learning by regression**; • **Applied computing** → *Economics*; • **Information systems** → *Data mining*.

Keywords

statistical arbitrage, cryptocurrency, cross-exchange spreads, gradient boosting, market microstructure, limit order book, live evaluation

CROSS-EXCHANGE SPREAD FORECAST → HIGH-CONFIDENCE TRADE

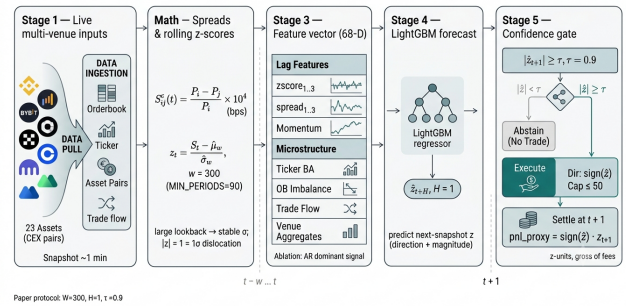


Figure 1: End-to-end paper protocol: live multi-venue inputs → cross-exchange spread and rolling z-score ($w=300$) → 68-D features (AR lags dominant) → LightGBM forecast $\hat{z}_{t+1}$ ($H=1$) → confidence gate $|\hat{z}| \geq \tau=0.9$ → selective trade and $t+1$ settlement in z-units.

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1 Introduction

Cryptocurrency markets are fragmented across dozens of centralized exchanges (CEXs), each with independent order books, heterogeneous fees, liquidity, and matching-engine latencies. Unlike equities, there is no consolidated tape that forces immediate price convergence. The result is recurrent cross-exchange price dislocations for identical assets [7]. These gaps present economically meaningful opportunities for same-asset statistical arbitrage, especially among high-volatility coins where short-lived Law-of-One-Price deviations appear at approximately one-minute granularity.

Existing high-frequency crypto pairs-trading systems largely treat the rolling z-score of a spread as a mechanical entry/exit rule: trade when $|z_t|$ exceeds a fixed threshold and exit on reversion [1, 5, 8, 12]. However, this design is not as reliable in the cross-exchange setting. At minute scale, spreads sometimes revert quickly and sometimes persist through funding or liquidity regimes; a fixed threshold always bets on mean reversion without understanding the broader context of the exchange dynamics at that moment. Venues also differ systematically in fees, depth, and latency, so a global rule treats every (coin, exchange-pair) identically, while missing potential systematic differences between exchanges. Checking z_t at every snapshot is cheap, but it leaves these two sources of structure

unused, which may in turn impact the quality of both the direction and magnitude of the short-horizon forecasts.

Here we replace thresholding the current z-score with a supervised forecast of the next one. We train a single LightGBM [4] panel model to predict $y_t = z_{t+H}$ (with $H=1$ in the production protocol) for same-asset cross-exchange spreads, conditioning on lagged spread dynamics, live L2 orderbook imbalance and trade flow, and native coin / venue-pair categoricals, as well as data about the trades, funding rate, and open interest. Bid/ask imbalance, which comes from analysing the L2 order book, is especially of interest to us in this study, as it encodes directional pressures and slippage risk that mechanical z-score rules can never access, and it cannot be reconstructed from historical OHLCV APIs. A confidence filter is implemented which then trades only when $|\hat{z}| \geq 0.9$, selecting high-magnitude z-score forecasts to trade on, rather than forcing an action on every snapshot, even when there is no arbitrage opportunity.

Methodologically, this is a change of formulation, not only of market setting. The prediction target is a local mean–standard-deviation normalization suited to short-horizon directional and magnitude learning; the model architecture learns venue- and asset-specific dynamics that global thresholds cannot represent; and finally the live-collected microstructure features define an evaluation setting that backtest-only studies cannot replicate. We enforce a chronological train/test split, score capacity-matched mechanical baselines on identical rows and settlement rules, and validate end-to-end in live paper-trading under the system’s real collection and decision latency, including multi-day campaigns at approximately one-minute granularity. We report on our model’s directional accuracy, R^2 , and pn1_proxy . The synchronized multi-exchange signal stream is released publicly through the public dataset release (URL withheld for double-blind review) as a standalone dataset contribution.

1.1 Contributions

- (1) **Learned next-snapshot z-score forecasting for same-asset cross-exchange spreads.** We replace mechanical $|z_t|$ thresholds with a LightGBM forecast of z_{t+H} , conditioned on live microstructure and coin/venue categoricals, with a $|\hat{z}| \geq 0.9$ confidence gate for selective entry.
- (2) **Live paper-trading under real decision latency.** We run the policy in live campaigns (including a ~ 72 -hour session) on direct centralized-exchange market data at approximately one-minute frequency: the trained LightGBM model predicts the next z-score at each snapshot and executes paper trades under the system’s real collection and decision delay.
- (3) **Clear gains over matched mechanical baselines on short and long live campaigns.** Scoring in z-units (not dollars), the paper gate $|\hat{z}| \geq 0.9$ on the ~ 73 -hour out-of-sample book yields about 87% directional wins and mean pn1_proxy of +1.37 on 12,795 closes; the Jul 31 ~ 8 -hour session shows the same pattern (DirAcc 84.9%; filtered R^2 of 0.51) and, under matched capacity at $\tau=0.9$, improves win rate by 15.4 percentage points and mean pn1_proxy

by 63% over capacity-matched z-score persistence under identical settlement rules.

- (4) **Public code, model, and microstructure dataset.** We release the repository, trained LightGBM model, and synchronized multi-exchange signals across 23 volatile assets and 6 exchanges (the code repository (URL withheld for double-blind review); the public dataset release (URL withheld for double-blind review)), including L2 orderbook and trade flow that cannot be reconstructed from standard historical exchange APIs, plus funding and open interest for downstream use.

2 Related Work

Classical cryptocurrency pairs trading largely focuses on methods used in statistical arbitrage and pairs trading in equity markets. They apply distance, cointegration, and z-score rules to single-exchange, different-asset pairs. Fil and Kristoufek [1] implement distance and cointegration methods to a crypto pairs trading setting, and find that they underperform classical benchmarks, while showing that performance is highly sensitive to parameters such as the collection frequency, and the execution window. Other studies present rule-based trading frameworks for crypto markets, [5, 8, 12, 13], however spread persistence between coins and across exchanges may vary with liquidity and funding regimes, venues differ in fees and latency, and high-volatility assets can produce frequent but short-lived price dislocations. Fixed thresholds cannot distinguish fast from slow reversion episodes, and they treat every venue pair identically. Therefore, we decided to explore same-asset cross-exchange opportunities, where the Law of One Price helps guarantee mean reversion, using a learned model, which can better exploit the different patterns and asymmetries present in the volatile crypto market. The target variable in our model is the forecast of the future rolling z-score, which informs whether or not we should trade on a certain asset.

A smaller literature trains models on spread or relative-return dynamics. Fischer et al. [2] show that machine-learning statistical arbitrage is feasible at minute scale in cryptocurrency markets, motivating our focus on a tighter same-asset cross-exchange target rather than cross-sectional coin ranks from lagged returns. They also study sensitivity to execution delay; they find that execution on a profitable trade needs to occur within less than five minutes from the detected arbitrage opportunity, or else the opportunity for profits disappears entirely. This finding helps motivate our decision to forecast the next z-score, as live trading entails execution delays from the time a transaction is initialized to when it is settled. Therefore, forecasting the next z-score and trading based on these predictions helps circumvent this practical challenge. We then take a further step by deploying the model in live paper-trading campaigns on real-time collected signals, so decisions are made under the system’s actual collection and decision latency and scored against subsequent market outcomes. Other research has deployed different models to the crypto arbitrage setting, including DNN/LSTM crypto spread models, equities LOB predictors, and abstention-based filters [3, 6, 9–11, 14]. These provide us with baselines to compare our model performance in the future, but any comparison between our results and theirs should be

interpreted with caution, as our research setting is unique among them: none of the previous work combines minute-scale same-asset cross-exchange pairs trading, so metrics about returns, Sharpe ratio, etc. are not a perfect one-to-one comparison.

Methodologically, the uniqueness of our research comes from both the model and the problem setting. The supervised prediction target is the future rolling z-score $y_t = z_{t+H}$ (local mean-standard-deviation normalization for short-horizon direction and magnitude) of the next time step relative to the observation at time t ; next, a confidence filter trades only when $|\hat{z}| \geq 0.9$. The model trained is a gradient-boosted tree model using LightGBM [4], which takes into account variables including coin and venue-pair categoricals, ticker, orderbook, trade, funding, and open interest data. We collected this data in real time, as it informs short-horizon movement and cannot be reconstructed from historical OHLCV APIs. This is available in our public multi-exchange signal release the public dataset release (URL withheld for double-blind review). We evaluate with directional accuracy and R^2 for forecast quality, and $pn1_proxy$ plus hourly portfolio Sharpe for trading stability, against capacity-matched mechanical baselines that threshold the current z-score at the same gate, plus an offline size-matched LSTM as a sequence-model control. Live campaigns, including a multi-day ~72-hour session at approximately one-minute granularity, test whether offline skill holds on real-time, non-replayable market state.

3 Methodology

3.1 Cross-Exchange Spread and Z-Score Target

Let $P_i^c(t)$ denote the mid-price of asset c on exchange i at snapshot t . For an exchange pair (i, j) , the cross-exchange spread in basis points is:

$$S_{ij}^c(t) = \frac{P_i^c(t) - P_j^c(t)}{P_i^c(t)} \times 10,000 \quad (1)$$

Unlike traditional pairs trading where the spread between *different* assets may diverge permanently as fundamentals change, the cross-exchange spread for the *same* asset is anchored by the Law of One Price: absent frictions, $S_{ij}^c(t) \rightarrow 0$. In practice, fees, withdrawal delays, and matching-engine latency allow transient deviations that are mean-reverting by construction.

We normalize the spread into a rolling z-score:

$$z_t = \frac{S_t - \hat{\mu}_w}{\hat{\sigma}_w} \quad (2)$$

where $\hat{\mu}_w$ and $\hat{\sigma}_w$ are the estimated rolling mean and standard deviation over a window of w snapshots, with a minimum period requirement before the z-score is defined. Model outputs $\hat{z}$ are therefore in units of this trailing $\hat{\sigma}_w$: $|\hat{z}| = 1$ is a one- σ forecasted displacement. We use $w = 300$ (MIN_PERIODS = 90); larger w adds context but lengthens warmup (Section 6).

3.2 Supervised Prediction Target

The model's target is the *future* z-score at horizon H :

$$y_t = z_{t+H} \quad (3)$$

computed via `groupby(coin, pair).zscore.shift(-H)` on the training panel. This is a regression target, not a binary classification. The model predicts both the direction and magnitude of the z-score H snapshots ahead, enabling a confidence filter on $|\hat{z}|$.

For the prediction model used in the paper-trading campaigns, $H = 1$ (predict the next snapshot's z-score), $w = 300$ snapshots, and MIN_PERIODS = 90. Window length and gate τ are justified in Section 6.

3.3 Feature Engineering

The feature matrix is constructed from four synchronized signal families collected at each snapshot: spread matrices, ticker quotes, L2 orderbook, and recent trades. Features are computed per (coin, exchange-pair, snapshot) observation.

3.3.1 Spread and z-score features. Lagged values of the spread in basis points and the rolling z-score form the base feature set:

- `spread_bps_lag{1, 2, 3}`: raw spread at lags 1–3.
- `zscore_lag{1, 2, 3}`: rolling z-score at lags 1–3.
- `spread_momentum_1_3`: difference between lag-1 and lag-3 spread, capturing short-term trend.
- `zscore_momentum_1_3`: same for z-scores.
- `zscore_accel`: second-order difference $z_{t-1} - 2z_{t-2} + z_{t-3}$, capturing acceleration in z-score trajectory.

3.3.2 Ticker microstructure features. From each exchange's best-bid/ask quotes:

- `tk_mid_lag{1, 2}`: lagged mid-prices per exchange.
- `tk_spread_bps_lag{1, 2}`: lagged bid-ask spreads in bps per exchange. Wider spreads indicate lower liquidity and higher execution cost.
- `tk_bid_volume_lag`, `tk_ask_volume_lag`: lagged top-of-book volumes per exchange.

3.3.3 Orderbook imbalance features. From the top 20 price levels on each side of each exchange's order book, we compute:

$$\text{imbalance}_i = \frac{V_{\text{bid},i} - V_{\text{ask},i}}{V_{\text{bid},i} + V_{\text{ask},i}} \quad (4)$$

where $V_{\text{bid},i}$ and $V_{\text{ask},i}$ are the total volumes on the bid and ask sides of exchange i 's top-20 levels. This ratio ranges from -1 (entirely ask-dominated) to $+1$ (entirely bid-dominated) and captures directional pressure in the order book. Lagged imbalance values `ob_imbalance_lag{1, 2, 3}` for each exchange provide the model with recent order flow dynamics.

3.3.4 Trade flow features. From the most recent trades on each exchange:

- `tr_buy_sell_ratio`: ratio of buy-initiated to sell-initiated trade volume, indicating whether recent execution flow is dominated by buyers or sellers.
- `tr_total_volume`: total trade volume, a proxy for activity and liquidity.

Both are lagged at 1–2 snapshots per exchange.

3.3.5 Cross-exchange aggregates. Features that capture the state of the multi-venue market as a whole:

- `cross_mid_range`, `cross_mid_std`: range and standard deviation of mid-prices across exchanges, measuring instantaneous cross-venue dispersion.
- `cross_ba_std`, `cross_ob_std`, `cross_ob_range`: cross-exchange variability in bid-ask spreads and orderbook imbalance.
- `net_ob_pressure`: aggregate orderbook imbalance across venues.
- `price_rank_{exchange}`: ordinal rank of each exchange's mid-price within the venue set, identifying which exchange is currently pricing high or low.
- `tightest_ba`, `widest_ba`, `ba_range`: spread-liquidity extremes across venues.

3.3.6 *Categorical identifiers.* `coin` and `pair` (exchange-pair label) are encoded as LightGBM native categoricals, allowing the model to learn asset-specific and venue-pair-specific dynamics without separate per-pair models.

The prediction model uses 68 features in total. OHLCV candle features are explicitly excluded (`LOAD_TABLES['ohlcv']=False`) because the live microstructure signals (ticker, orderbook, trades) provide strictly more information at the snapshot level.

3.4 Model: LightGBM Gradient Boosting

We train a single LightGBM [4] regression model on the pooled panel of all (coin, exchange-pair, snapshot) observations. The objective is squared error minimization on $y_t = z_{t+H}$.

Table 1: LightGBM hyperparameters for the prediction model.

Parameter	Value
Max boosting rounds	2,500
Early stopping patience	250 rounds
Best iteration (selected)	74
Number of leaves	255
Learning rate	0.1
Min child samples	200
Feature fraction	0.6
Bagging fraction	0.7
L1 regularization	0.1
L2 regularization	0.1

Early stopping selects iteration 74 out of 2,500 maximum rounds, indicating that the model converges rapidly and additional boosting does not improve validation loss. The large minimum child sample (200) and sub-unity feature/bagging fractions provide regularization against overfitting to the dependent cross-sectional structure of the panel.

3.5 Trading Policy

At each live snapshot, the model generates a prediction $\hat{z}_{t+1}$ for every (coin, exchange-pair) observation. The trading policy is:

- (1) **Entry:** open a position if $|\hat{z}_{t+1}| \geq \tau$ and the (coin, pair) slot is not already occupied. The paper protocol uses $\tau=0.9$ (Section 6).

- (2) **Direction:** $\text{sign}(\hat{z}_{t+1})$. The model predicts whether the spread z-score will be positive or negative at $t+1$ and bets accordingly.
- (3) **Capacity:** at most `max_open` = 50 positions open simultaneously, enforced by a first-come queue.
- (4) **Settlement:** each position settles at $t+H$ (one snapshot later for $H=1$). The proxy profit is $\text{pnl_proxy} = \text{direction} \times z_{t+H}$.

The settlement proxy measures whether the z-score moved in the predicted direction and by how much, in z-score units. It is not a dollar profit metric and does not account for transaction fees or slippage.

3.6 Evaluation Metrics

We report both forecast quality and trading performance metrics:

- **R^2** (coefficient of determination): fraction of variance in the realized z_{t+H} explained by the model's predictions. Reported on all predictions and on the filtered subset.
- **Directional accuracy (DirAcc):** fraction of predictions where $\text{sign}(\hat{z}) = \text{sign}(z_{t+H})$.
- **Mean pnl_proxy:** average $\text{direction} \times z_{t+H}$ across settled trades.
- **Per-trade Sharpe:** $\text{mean}(\text{pnl_proxy}) / \text{std}(\text{pnl_proxy})$ on the filtered trade set ($R_f = 0$), used to select τ in Section 6.
- **Hourly portfolio Sharpe ratio:** $\text{mean}(\Delta \text{equity}_h) / \text{std}(\Delta \text{equity}_h)$, where hourly equity includes realized closed-trade profits plus mark-to-market valuation of open positions as $\text{direction} \times z_t$. Risk-free rate is zero because the z-unit proxy has no currency denomination and represents forecast quality rather than economic returns.

All metrics are computed with a matched **naive persistence baseline** ($\hat{z}_{t+H} \leftarrow z_t$) scored on the identical row set, ensuring that gains from the confidence filter cannot be mistaken for model skill.

4 Experimental Setup

4.1 Data Collection

We collect live microstructure data from centralized cryptocurrency exchanges using a custom collector that polls 9 signal families (orderbook, trades, ticker, funding, open interest, withdrawal status, exchange status, spread matrices, OHLCV) via REST API at approximately 60-second target cadence across 23 assets and up to 6 exchanges (Binance, Bybit, OKX, Coinbase, Kraken, MEXC). Under full signal load, the observed cadence is approximately 110 seconds per snapshot. Data collection protocol and signal schema are documented in the repository accompanying the public dataset release (URL withheld for double-blind review).

4.2 Asset Universe

The 23-asset universe was selected for high cross-exchange spread volatility and sufficient liquidity across multiple venues, requiring listing on at least 3 of the 6 target exchanges. See the repository for the full asset list.

4.3 Train/Test Split

All collection windows before July 25, 2026 form the training set; July 25–28 forms the held-out test set. The split is strictly chronological with no shuffling. The training set contains approximately 2.9 million (coin, exchange-pair, snapshot) feature rows spanning multiple collection windows from June–July 2026. Cross-sectional rows within a snapshot are structurally dependent (they share the same market state), so this row count should not be interpreted as independent samples.

4.4 Live Paper-Trading Protocol

For live validation, we run independent paper-trading campaigns that process incoming collector data in real time with no access to future snapshots. Capacity is $\text{max_open} = 50$. The paper confidence gate is $\tau=0.9$ (Section 6).

4.4.1 Campaign A: Jul 31.

- **Model:** 68-feature prediction model (statarb_lgbm.txt).
- **Horizon:** $H = 1$ (predict next snapshot).
- **Z-score window:** 300 snapshots.
- **Duration:** approximately 8 hours of useful data (collector ran for 264 snapshots).
- **Warmup:** predictions become active after $\text{MIN_PERIODS} = 90$ snapshots (approximately 2.5–3 hours).
- **Predictions scored:** 58,995 total; 2,026 rows with $|\hat{z}| \geq 0.9$.

4.4.2 Campaign B: Aug 4–7 (~72h).

- **Model:** same 68-feature, $H=1$, $w=300$ policy as Campaign A.
- **Duration:** ≈ 72.9 hours of active trading (50,690 closed trades; 1.01M scored predictions).
- **Role:** multi-day live stress test; primary $\tau=0.9$ metrics are the post-hoc paper-protocol subset of this book (Section 6).

Campaign A is a short live stress test; Campaign B extends the same live deployment settings over three days. We present them as complementary live evidence rather than a controlled factorial experiment.

4.5 Baseline Definitions

4.5.1 Naive persistence. On every row where the model generates a prediction, we also record the naive forecast $\hat{z}_{t+H} \leftarrow z_t$ (the current z-score predicts the future z-score unchanged). R^2 and directional accuracy are computed for both model and naive on the *identical* rows, ensuring that any performance difference is attributable to the model rather than to row selection.

4.5.2 Mechanical z-score baselines. We replay two mechanical trading rules on the Jul 31 signal panel under the same capacity constraints ($\text{max_open} = 50$), settlement proxy, and paper gate $\tau=0.9$ as the model:

- **Persistence:** enter when $|z_t| \geq 0.9$, direction = $\text{sign}(z_t)$.
- **Mean-reversion:** enter when $|z_t| \geq 0.9$, direction = $-\text{sign}(z_t)$.

Scarce slots are filled by ranking $|\hat{z}|$ (model) or $|z_t|$ (mechanical). These represent the two canonical directions a rule-based z-score system can take: betting the spread continues (persistence) or reverts (classical pairs trading). Both settle at $t+1$ with $\text{pnl_proxy} = \text{direction} \times z_{t+1}$, identical to the model's settlement.

4.6 Reproducibility

All training, evaluation, and baseline scripts are available in the code repository (the code repository (URL withheld for double-blind review)), and the dataset is released at the public dataset release (URL withheld for double-blind review). LightGBM training is reproducible given the hyperparameters in Table 1.

5 Results

We evaluate the LightGBM z-score forecasting model across offline held-out evaluation and two live paper-trading settings (Jul 31 and a ~72h Aug 4–7 campaign). In each setting we report both forecast quality metrics (R^2 , directional accuracy) and trading performance metrics (mean profit proxy, hourly portfolio Sharpe ratio), always scoring a naive persistence baseline on the same rows as the model. The paper confidence gate is $\tau=0.9$ (justified in Section 6).

5.1 Offline Evaluation

The model is trained on all collection windows before July 25, 2026 and tested on the Jul 25–28 held-out window. Table 2 reports forecast metrics on the full prediction set and on the paper-protocol filter $|\hat{z}| \geq 0.9$.

Table 2: Offline evaluation on the Jul 25–28 held-out test set. Filtered rows use the paper gate $|\hat{z}| \geq 0.9$.

Set	R^2	DirAcc
Test (all predictions)	0.133	62.8%
Test ($ \hat{z} \geq 0.9$)	0.535	85.3%
Train (all predictions)	0.128	61.8%

Train and test R^2 are nearly identical on the unfiltered set (0.128 vs 0.133), indicating no overfitting (early stopping selected iteration 74 of 2,500). Under $\tau=0.9$, offline test DirAcc rises to 85.3% and R^2 to 0.535. Live $\tau=0.9$ subsets (Section 6) remain consistent with this filtered offline skill.

5.2 Live Paper-Trading Campaigns

5.2.1 Campaign A (Jul 31). The production 68-feature model with horizon $H = 1$ and z-score window 300 was deployed for an 8-hour session. Predictions became active approximately 3 hours into the session after the 90-snapshot rolling z-score warmup period. Table 3 reports model and naive persistence metrics on all scored rows and on the paper gate $|\hat{z}| \geq 0.9$.

Table 3: Jul 31 live campaign: model vs naive persistence (paper gate $|\hat{z}| \geq 0.9$).

Set	n	Model R^2	Model Dir.	Naive R^2	Naive Dir.
All	58,995	0.104	60.9%	-0.338	65.6%
$ \hat{z} \geq 0.9$	2,026	0.514	84.9%	0.424	84.4%

On Jul 31 the model achieves positive R^2 where naive persistence is negative or near zero on the full panel. Under $|\hat{z}| \geq 0.9$, naive directional accuracy nearly matches the model (84.4% vs 84.9%),

but the model's R^2 advantage (0.514 vs 0.424) shows that it captures the *magnitude* of z-score movements, not merely their sign.

5.3 Baseline Comparison: Learned Forecast vs Mechanical Z-Score Rules

To isolate the value of the learned forecast from the confidence filter, we replay two mechanical baselines on the Jul 31 signal panel under matched capacity ($\text{max_open} = 50$) and the same magnitude gate $\tau=0.9$. Both baselines enter when $|z_t| \geq 0.9$ but use the current z-score rather than a predicted future z-score to determine trade direction; scarce slots are filled by ranking $|z_t|$ (LightGBM ranks $|\hat{z}|$):

- **Persistence:** direction = $\text{sign}(z_t)$. Bets that the current z-score will continue into the next snapshot.
- **Mean-reversion:** direction = $-\text{sign}(z_t)$. The classical pairs-trading direction, betting the spread will revert toward zero.

Table 4 reports the capacity-matched baseline comparison, which is the primary evidence for the model's improvement over rule-based z-score thresholds.

Table 4: Capacity-matched baseline comparison on Jul 31 at paper gate $\tau=0.9$ ($\text{max_open} = 50$, ranked fill). Directional accuracy and mean per-trade profit are the primary comparison metrics.

Strategy	Direction	n closed	DirAcc	Mean PnL
LightGBM	$\text{sign}(\hat{z})$	2,026	84.9%	+1.195
Mech. persistence	$\text{sign}(z_t)$	8,550	69.5%	+0.731
Mech. mean-reversion	$-\text{sign}(z_t)$	8,550	30.5%	-0.731

As shown in Table 4, the learned policy improves directional accuracy by 15.4 percentage points and mean per-trade profit by 63% over mechanical persistence under the same $\tau=0.9$ gate and capacity. Classical mean-reversion produces negative z-unit returns at this hold horizon. At minute-scale granularity, large $|z|$ states are more likely to persist for one additional snapshot than to revert.

Figure 2 visualizes the directional accuracy and mean per-trade profit for all three policies. Figure 3 shows the cumulative z-unit profit trajectories over the Jul 31 session.

5.4 Portfolio Risk-Adjusted Performance

Over the 6 active hours of the Jul 31 $\tau=0.9$ ranked-capacity book, the model's hourly portfolio Sharpe ratio is 1.90 (mean hourly z-unit profit divided by its standard deviation, including mark-to-market).

5.5 Campaign B (~72h) under the Paper Gate

Campaign B extends the Campaign A settings ($H=1$, $w=300$, $\text{max_open}=50$) over ≈ 72.9 hours. Under the paper protocol $|\hat{z}| \geq 0.9$ applied to settled closes, the book has $n=12,795$ trades with DirAcc 86.7%, R^2 0.599, mean z-proxy +1.37, and per-trade Sharpe 1.08 (Table 5)—consistent with the Jul 31 fine-sweep operating point and confirming that the short session was not a one-off.

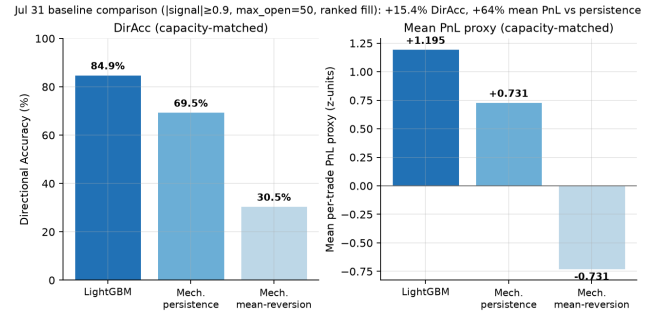


Figure 2: Capacity-matched baseline comparison on Jul 31 at $\tau=0.9$. The learned policy improves directional accuracy by 15.4 pp and mean per-trade profit by 63% over mechanical persistence. Mean-reversion is negative on both metrics.

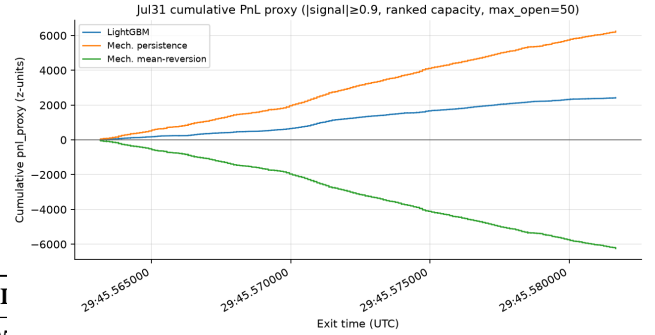


Figure 3: Cumulative z-unit proxy P&L over the Jul 31 session at $\tau=0.9$ (ranked capacity). This is a proxy in z-score units gross of costs, not a currency return; see Section 7.1.

6 Ablation Studies

6.1 Selecting Window Length and Confidence Threshold

Window w . Increasing w enriches $(\hat{\mu}_w, \hat{\sigma}_w)$ but forces longer warmup before a valid z exists and slows σ adaptation. Offline spreads-only LightGBM sweeps on the Jul 25 holdout show test skill rising into a plateau near $w \approx 560-720$, with only modest gains past the production choice $w=300$ ($\approx +1$ pp DirAcc and $+0.017 R^2$ at $w=720$ for $\sim 2.4\times$ lookback). A $w \times \tau$ grid scored on the Jul 31 collector likewise does not favor larger w at matched τ . We therefore retain $w=300$.

Threshold τ . Early live collection used a permissive gate $\tau=0.5$ to maximize coverage under $\text{max_open}=50$. That choice is a starting point for selection, not the paper operating point: on Jul 31 signal rows it yields DirAcc 76.7%, R^2 0.347, mean z-proxy +0.75, and per-trade Sharpe only 0.67 ($n=8,775$). A fine sweep on the same panel shows that raising τ monotonically improves DirAcc, R^2 , mean z-proxy, and per-trade Sharpe mean(pnl_proxy)/std(pnl_proxy) ($R_f = 0$). **Per-trade** Sharpe first reaches ≥ 1 at $\tau=0.9$ (Table 5): DirAcc 84.9%, R^2 0.514, mean z-proxy +1.20, Sharpe 1.02 on $n=2,026$ rows—still a large book relative to $\tau=1.0$ ($n=1,426$), with mean pnl_proxy already high versus $\tau=0.5$. Neighboring points bound

the choice: $\tau=0.85$ remains below Sharpe 1, while $\tau=1.0$ adds little Sharpe for another $\sim 30\%$ coverage cut. The same $\tau=0.9$ cut on Campaign B closes confirms the lift from the initial $\tau=0.5$ book at longer horizon (50,690 $\rightarrow$ 12,795; DirAcc 79.0% $\rightarrow$ 86.7%; Sharpe/trade 0.75 $\rightarrow$ 1.08). We therefore fix $\tau=0.9$ as the paper protocol; headline results use this gate (Section 7.3 notes when live fills are post-hoc filtered).

Table 5: Choosing τ from the initial live gate $\tau=0.5$ ($w=300$). Jul 31 rows are scored signals; Campaign B rows are live closes. Per-trade Sharpe uses $R_f = 0$.

Book	τ	n	DirAcc	R^2	Mean	Sharpe/tr
Jul 31 signals	0.5	8,775	76.7%	0.347	+0.75	0.67
Jul 31 signals	0.85	2,452	84.5%	0.497	+1.16	0.99
Jul 31 signals	0.9	2,026	84.9%	0.514	+1.20	1.02
Jul 31 signals	1.0	1,426	85.3%	0.534	+1.26	1.07
$\sim 72h$ closes	0.5	50,690	79.0%	0.439	+0.84	0.75
$\sim 72h$ closes	0.9	12,795	86.7%	0.599	+1.37	1.08

6.2 Feature Importance

Figure 4 shows the top 10 of the model’s 68 features ranked by LightGBM split gain.

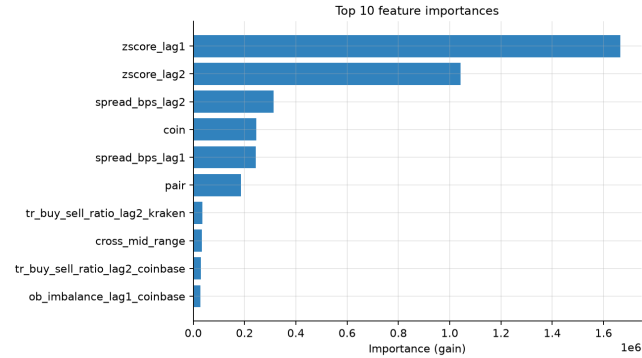


Figure 4: Top 10 features by LightGBM split gain.

The model mainly relies on z-score and spread lags. This semi-autoregressive structure makes sense given the problem context, wherein the previous states are strong indicators of future motion of the spread and z-scores. The coin and pair categorical variables are also included in the top 10, indicating that the model learns to differentiate between different coins on different exchanges. Finally, the microstructure features, namely trade data and L2 order book data, are also included in the top 10. Even though they are of smaller importance, these features provide some insight to the model.

6.3 Ablation: Pre-Training Feature Engineering vs. Raw Market Features

The prediction target is $y_t = z_{t+1}$ ($H=1$). We ask whether *transforming* the same live microstructure—mid prices into spreads, rolling z-scores, lags, and short momentum/acceleration—improves filtered skill relative to training on raw live columns alone.

Table 6: Engineered vs non-engineered microstructure features (Jul 25–28 offline, paper gate $|\hat{z}| \geq 0.9$). Identity (coin/pair) held fixed. Δ vs raw (non-engineered) columns: DirAcc_f +7.6 pp, R_f^2 +0.523.

Variant	R_f^2	ΔR_f^2	DirAcc _f	Δ DirAcc _f
Raw + identity	0.012	—	77.8%	—
Engineered + identity	0.535	+0.523	85.4%	+7.6 pp

Engineered trajectory. From venue mids we form pairwise spreads S (bps) and rolling z-scores z , then construct lagged S and z , plus short momentum and acceleration of those series (9 columns). This is a pre-training transform of collected market data, not a change to what the collector observes.

Raw market features. Ticker mids and bid–ask widths, L2 imbalance, trade volumes, and cross-venue aggregates (51 surviving columns)—the same session’s live signals without that trajectory transform.

Identity controls. coin and pair are held fixed on both sides of the comparison. They are not the engineering question: coin selects which asset’s dynamics apply; pair selects which two venues the cross-exchange dislocation (and mean reversion) lives on.

We retrain LightGBM on the Jul 25–28 offline holdout under fixed published LGBM_PARAMS and a shared boosting-round budget (49 rounds). Filtered metrics use $|\hat{z}| \geq 0.9$ (Table 5).

Result. The gain comes from transforming the initial microstructure, not from seeing different data. Compared to non-engineered microstructure features (raw ticker/OB/trade/cross columns with identity fixed), the mid $\rightarrow$ spread $\rightarrow$ z trajectory improves filtered directional accuracy by +7.6 pp (77.8% $\rightarrow$ 85.4%) and filtered R^2 by +0.523 (0.012 $\rightarrow$ 0.535). Under the paper gate, pre-training feature engineering on live mids is what moves next-snapshot z prediction, and the same lift carries through to higher mean pnl_proxy and per-trade Sharpe.

7 Discussion

7.1 Z-Score Units Capture Forecast Quality

The settlement proxy $\text{pnl}_{\text{proxy}} = \text{direction} \times z_{t+H}$ measures the model’s ability to correctly predict the standardized spread level at horizon H . It is designed as a forecast quality metric, not as an economic P&L.

When the same Jul 31 $\tau=0.9$ ranked-capacity trades are re-scored in basis points of realized spread change, the ordering between policies inverts: the learned policy averages -0.94 bps per trade (34.9% win rate), while mechanical mean-reversion yields +3.97 bps (76.0% win rate). Table 7 summarizes this difference.

This inversion reflects the model’s training objective: predicting the z-score level z_{t+H} (dominated by the slow-moving rolling mean) rather than the instantaneous spread change ΔS_t . The z-unit settlement is appropriate for evaluating forecast quality, which is the primary claim of this work. For production deployment, a direct spread-change target net of fees would align training with the economic objective.

Table 7: Jul 31 $\tau=0.9$ ranked-capacity entries scored in z-units and basis points of realized spread change, gross of fees. Policy ordering inverts between units.

Policy	Mean z-proxy	Mean gross (bps)	Gross win rate
LightGBM	+1.195	−0.94	34.9%
Mech. persistence	+0.731	−3.97	9.9%
Mech. mean-reversion	−0.731	+3.97	76.0%

7.2 Portfolio Sharpe Characterizes Forecast Stability

The Jul 31 hourly Sharpe of 1.90 ($n = 6$) under the $\tau=0.9$ ranked-capacity book characterizes forecast stability but is not directly comparable to the multi-month, cost-aware benchmarks in the crypto pairs-trading literature [12, 13]. The capacity-matched mechanical persistence peer attains a higher hourly Sharpe (2.62) while trailing on per-trade DirAcc and mean z-proxy, confirming that both policies exploit a high-persistence regime. Per-trade forecast quality (R^2 and directional accuracy) is the primary evidence for the model’s advantage; Sharpe is a secondary stability indicator.

7.3 Threats to Validity

Evaluation horizon. Campaign A covers approximately six active hours—too short alone to characterize regime dependence—but Campaign B extends live collection over ≈ 73 hours. Headline metrics use the paper gate $\tau=0.9$ (Section 6); some live $\tau=0.9$ numbers are post-hoc filters or ranked-capacity replays on the collected sessions rather than a separate live re-fill at that gate. The campaigns still differ in calendar window, so they are complementary stress tests rather than a controlled factorial ablation.

Proxy settlement. P&L is measured in z-units gross of fees, slippage, and bid-ask crossing (Section 7.1). No dollar return is claimed.

Execution latency. The collector’s observed cadence is approximately 110 seconds, so signals are acted on against market states up to two minutes stale. Fischer et al. [2] show that minute-scale crypto alpha can vanish within 1–5 minutes of delay; we have not run the corresponding ablation.

Dependence structure. Rows within a snapshot share market state, so the 2.9 million training rows and 59,000 predictions are not independent samples. We report no significance tests.

8 Conclusion

We presented a gradient-boosting framework that replaces mechanical z-score thresholds with a learned forecast for cross-exchange cryptocurrency spread dynamics. Under the paper gate $|\hat{z}| \geq 0.9$ —chosen as the first operating point with per-trade Sharpe ≥ 1 while keeping high mean z-proxy without over-thinning the book (Section 6)—offline holdout DirAcc reaches 85.3% (R^2 0.535), Jul 31 $\tau=0.9$ entries reach DirAcc 84.9% (R^2 0.514), and the ~ 72 h campaign subset reaches DirAcc 86.7% (mean z-proxy +1.37). Under matched capacity at the same gate, the learned policy improves per-trade accuracy by 15.4 pp and mean z-unit profit by 63% over mechanical

persistence. Critically, this is a forecasting result: scored in basis points of realized spread change, none of the policies—including ours—has a positive mean at any non-zero transaction cost (Section 7.1). Section 7.3 details the threats to validity.

8.1 Future Work

Incorporating realistic fee structures, execution latency, and a direct spread-change training target would bridge the gap between z-unit performance and economic profitability. The released dataset enables the community to benchmark alternative architectures—including sequence models that may capture temporal dependencies the tree ensemble cannot—on the same cross-exchange microstructure panel.

Ethics and Privacy Statement

This study uses only publicly available aggregate market data (quotes, orderbook depth, anonymous trade prints, funding rates) from public REST endpoints of centralized cryptocurrency exchanges. No personal data or account identifiers are present; no human subjects were involved. Data collection respected each venue’s published rate limits; no authenticated or access-restricted endpoints were used.

All trading is simulated—no orders were submitted, no capital was deployed, and the reported profits are proxy quantities in z-score units. As documented in Section 7.1, the strategies evaluated here are *not* profitable once realistic transaction costs are applied. The accompanying dataset contains only the public market observations described above.

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